MILITARY SPECIFICATION SHEET

ELECTRON TUBE, CATHODE RAY

TYPE 10KP7A

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

DESCRIPTION: Magnetic deflection and focus

PIN CONNECTIONS AND DIMENSIONS: See figure 1

ABSOLUTE RATINGS:

Parameter:	Ef	Ec1	Ec2	Eb	Ehk	Rg1	Alt
Unit:	V	Vdc	Vdc	Vdc	Vdc	Meg	ft
Maximum:	6.9	0, -180	770	11,000	±125	Note 1	60,000
Minimum:	5.7	---	---	7,000	---	---	---
		Note 1					
TEST CONDITIONS:	6.3	Adjust	250	9,000	---	---	---

GENERAL:

Qualification - Required

© denotes changes

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T14:14Z
Check the source to verify that this is the current version before use.

METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	SYMBOL	LIMITS		UNIT
					MINIMUM	MAXIMUM	
	Qualification inspection						
1216	Base material insulating quality	-	Zone 5 (min)	---	---	---	---
5216	Cathode illumination	-		---	---	---	---
1002	Barometric pressure, reduced	2	54 mmHg	---	---	---	---
1141	Pressure (implosion)	-		---	---	---	---
1331	Direct-interelectrode capacitance	-	k to all g1 to all g2 to all	Ck Cg1 Cg2	--- --- ---	8.0 10.0 10.0	pF pF pF
5111	Vibration (cathode-ray tubes)	3		Width	---	2.0	mm
(C) 5115	Shock	-		---	---	---	---
	Quality conformance inspection, part 1						
5201	Voltage breakdown	-		---	---	---	---
5201	Voltage breakdown (magnetic types)	-		---	---	---	---
5106	Bulb, screen, and faceplate quality	-		---	---	---	---
5223	Modulation	-	Ib = 200 μAdc	ΔEc1	---	38	Vdc
5231	Spot position (magnetic deflection)	-		Distance	---	18	mm
5241	Grid cutoff voltage	-		Ec1	-27	-63	Vdc
5251	Grid No. 1 leakage current	-		---	---	---	---
5236	Zero-bias anode current (magnetic deflection)	-		---	---	---	---
5206	Gas ratio	4		Gr	---	0.25	---
5101	Neck straightness	5		---	---	---	---
---	Aperture alinement	6		Distance	---	8.0	mm
	Quality conformance inspection, part 2						
1301	Heater current	-		If	540	660	mA
5221	Screens	7		---	---	---	---
5251	Heater-cathode leakage current	-		---	---	---	---
1101	Secureness of base, cap, or insert	-		---	---	---	---

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T14:14Z
Check the source to verify that this is the current version before use.

METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	SYMBOL	LIMITS		UNIT
					MINIMUM	MAXIMUM	
	Quality conformance inspection, part 2 -Continued						
5216	Stray emission (conventional types)	-	Eb = 11,000 Vdc; Ec2 = 770 Vdc	---	---	---	---
5101	Side terminal and base alinement	-		---	---	---	---
5226	Line width A (magnetic deflection)	-	Ib = 200 μAdc	Width	---	0.38	mm
5226	Line width C (magnetic deflection	-	Ib = 200 μAdc	Width	---	0.55	mm
5201	Electrode currents (grid No. 2)	-	Ec1 = 0	Ic2	---	15	μAdc
5246	Focusing ampere turns (magnetic deflection)	-	Ib = 200 μAdc; D = 3-1/4 inches (82.55 mm)	Ampere turns	450	570	---
5101	Neck and bulb alinement	-		---	---	---	---
5251	Grid No. 2 leakage current	-		---	---	---	---
5101	Face tilt	-		---	---	---	---
Ⓒ 1111	Base pin solder depth	-		---	---	---	---
1105	Permanence of marking	-		---	---	---	---
	Quality conformance inspection, part 3						
---	Life test	-	Group C; Eb = 11,000 Vdc; Ec2 = 770 Vdc; Ib = 60 μAdc; t = 500 hours (min)	---	---	---	---
---	Life-test end points:						
5226	Line width A	-		Width	---	0.38	mm
5226	Line width C	-		Width	---	0.55	mm
5223	Modulation	-		ΔEc1	---	38	Vdc
5251	Heater-cathode leakage current	-		---	---	---	---
5251	Grid No. 1 leakage current	-		---	---	---	---
5251	Grid No. 2 leakage current	-		---	---	---	---
5216	Stray emission (conventional types)	-		---	---	---	---

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T14:14Z
Check the source to verify that this is the current version before use.

NOTES:

1. When Ec2 is greater than 330 Vdc or Ec1 is near zero, or both, the effective resistance of the anode and grid No. 3 supply should be adequate to limit the anode and grid No. 3 input power to 6 watts. The peak grid No. 1 drive from cutoff should never exceed 65 volts.

2. Maximum-rated voltages are applied to all electrodes connected through the base, and with tube in cutoff conditions.

3. Displacement of the spot corresponding to the image of the final aperture shall not exceed the limit specified.

4. This test to be performed at the conclusion of the holding period.

5. The neck and base straightness shall be determined by the insertion of the tube neck in a cylinder 5-inches (127.0 mm) long and $1.500\ ^{+0.003}_{-0.000}$ inches ($38.10\ ^{+0.08}_{-0.00}$ mm) inside diameter. The cylinder should move freely between the reference line and the base of the assembled tube.

6. The distance between the center of the unfocused, undeflected spot at low intensity (Ec1 near cutoff) and the center of the image of the masking aperture observed at high intensity of the unfocused, undeflected spot shall not exceed the limit specified herein. To prevent damage to the screen, Ec1 should not be held at zero for more than approximately 10 seconds.

7. The screen characteristics shall be measured with constant beam energy of 0.24 watts defocused to a spot approximately 0.25 cm in diameter. The test conditions shall be anode voltage (relative to cathode) 4,000 V minimum and beam current 60 μAdc maximum. The screen characteristics shall comply with the following minimum limits: cb5 = 400 cb; G5: 1 = 4.

Custodians:
Army - EL
Navy - EC
Air Force - 80

Review activities:
Air Force - 11, 17
DSA - ES
Army - MI, MU

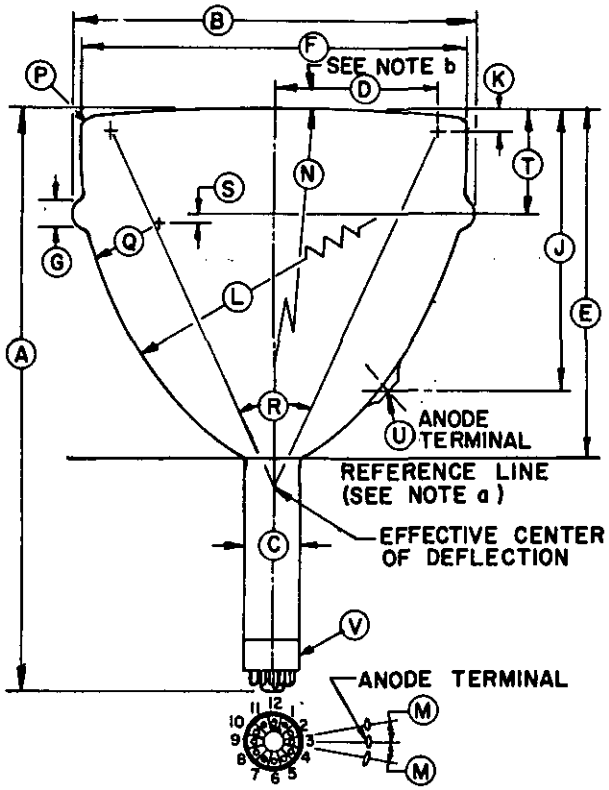
User activities:
Army - MC
Navy - AS, OS, MC, CG, SH

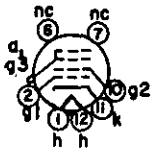
Preparing activity: Navy - EC

Agent: DSA - ES

(Project 5960-2603)

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T14:14Z
Check the source to verify that this is the current version before use.





PIN CONNECTIONS

Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
Qualification inspection		
U	Bulb contact: J1-21	
V	Base: B5-57 or B7-51	
Quality conformance inspection, part 2		
A	17.245(438.02)	17.995(457.07)
B		10.620(269.75)
C	1.380 (35.05)	1.500 (38.10)
D	4.500(114.30)	
E	9.250(234.95)	9.620(244.35)
F	10.380(263.65)	10.620(269.75)
G		.750 (19.05)
J	6.870(174.50)	7.620(193.55)
M		10°
Reference dimensions (see note c)		
K	.650 (16.51)	
L	20.000(508.00)	
N	42.000(1066.80)	
P	.380 (9.65)	
Q	2.000 (50.80)	
R	50°	
S	.242 (6.15)	
T	2.900 (73.66)	

NOTES:
a. Reference line is determined by position where JEDEC gage G-112 will rest on bulb cone.
b. Useful screen radius.
c. Reference dimensions are for information only and are not required for inspection purposes.

FIGURE 1. Outline drawing of electron tube type 10KP7A.

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T14:14Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T14:14Z
Check the source to verify that this is the current version before use.